



Advanced Time Series Analysis and Forecasting

Chapter 6: State space models and Bayesian filtering



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Acronyms used in this material (1/3)

- ▣ AI: Artificial Intelligence
- ▣ APF: Auxiliary Particle Filter
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BN: Beveridge–Nelson (decomposition)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BSTS: Bayesian Structural Time Series
- ▣ CLT: Central Limit Theorem
- ▣ CMRMTSPL: Real manufacturing and trade industries sales (FRED series code)
- ▣ DFM: Dynamic Factor Model
- ▣ DK: Durbin–Koopman (simulation smoother, 2002)
- ▣ DSGE: Dynamic Stochastic General Equilibrium (model)
- ▣ EKF: Extended Kalman Filter
- ▣ EM: Expectation–Maximisation (algorithm)
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ETS: Error, Trend, Seasonality (exponential smoothing state space models)
- ▣ FERMIAC: mechanical Monte Carlo device of Enrico Fermi (Los Alamos, 1947)
- ▣ FFBS: Forward Filtering, Backward Sampling
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GDP: Gross Domestic Product
- ▣ GDPC1: Real Gross Domestic Product (FRED code)
- ▣ GDPDEF: GDP implicit price deflator (FRED series code)

Acronyms used in this material (2/3)

- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ HP: Hodrick–Prescott (filter)
- ▣ IG: inverse gamma (distribution)
- ▣ IMA: Integrated Moving Average (model)
- ▣ INDPRO: Industrial production index (FRED series code)
- ▣ KSC: Kim–Shephard–Chib (1998), mixture sampler for stochastic volatility
- ▣ LLM: Large Language Model
- ▣ LR: Likelihood Ratio (test)
- ▣ MA: Moving Average
- ▣ MCMC: Markov Chain Monte Carlo
- ▣ ML: Machine Learning
- ▣ MNZ: Morley–Nelson–Zivot (2003)
- ▣ OLS: Ordinary Least Squares
- ▣ PAYEMS: Total nonfarm payroll employment (FRED series code)
- ▣ PMMH: Particle Marginal Metropolis–Hastings
- ▣ QML: Quasi-Maximum Likelihood
- ▣ SMC: Sequential Monte Carlo
- ▣ SV: Stochastic Volatility
- ▣ TSA: Time Series Analysis
- ▣ TVP: Time-Varying Parameter(s)
- ▣ TVP-VAR: Time-Varying-Parameter Vector AutoRegression
- ▣ TVP-VAR-SV: TVP-VAR with Stochastic Volatility
- ▣ UC: Unobserved Components (model)
- ▣ UC-SV: Unobserved Components model with Stochastic Volatility (Stock–Watson 2007)
- ▣ UC-UR: UC model with correlated trend and cycle shocks (Morley–Nelson–Zivot 2003)
- ▣ UC0: UC model with uncorrelated trend and cycle shocks (Clark 1987)

Acronyms used in this material (3/3)

- ▣ UKF: Unscented Kalman Filter
- ▣ US: United States
- ▣ VAR: Vector AutoRegression
- ▣ VAT: Value Added Tax
- ▣ W875RX1: Real personal income excluding current transfer receipts (FRED series code)

Today's question and route

- ▣ **Question:** how do we learn about quantities we never observe (trend inflation, the output gap, volatility, time-varying coefficients) and how much should we trust what we learn?
 - ▶ one framework: a state equation for the hidden quantity, a measurement equation for the data, and a filter that updates beliefs as data arrive
- ▣ **Route** of the chapter
 - ▶ the general linear Gaussian model, exact diffuse initialisation, the exact likelihood and inference at the boundary
 - ▶ Bayesian state space: simulation smoothers and Gibbs sampling; stochastic volatility
 - ▶ nonlinear filters: extended and unscented Kalman filters, particle filters, particle MCMC
 - ▶ time-varying parameters, dynamic factors, trend-cycle decompositions and trend inflation with stochastic volatility
- ▣ We build on [TSA, Chapter 10](#) (state space form, Kalman filter by hand, smoothing, local level and trend, the UC output gap, Markov switching basics); Seminar 6 comes before this lecture

Learning outcomes

- ▣ Derive the Kalman filter and smoother for the general linear Gaussian model, with exact diffuse initialisation, and write them in `numpy`
- ▣ Maximise the exact likelihood and judge inference on variances that may sit on the boundary (pile-up, bootstrap LR tests)
- ▣ Sample states with the Carter–Kohn, Durbin–Koopman and precision samplers inside a Gibbs sampler; estimate stochastic volatility by the KSC mixture
- ▣ Run extended, unscented and particle filters; use the particle likelihood in PMMH and choose the number of particles
- ▣ Estimate TVP regressions, UC models with correlated shocks and UC-SV trend inflation, and report how fragile the latent estimates are

Reading, data and tools

- Backbone: [Durbin and Koopman \(2012\)](#), Ch. 2–9 and 11–14; [Harvey \(1990\)](#); [Kim and Nelson \(1999\)](#); [Hamilton \(1994\)](#), Ch. 13; [Chopin and Papaspiliopoulos \(2020\)](#)
 - ▶ original papers: [Kalman \(1960\)](#), [Koopman \(1997\)](#), [Carter and Kohn \(1994\)](#), [Durbin and Koopman \(2002\)](#), [Kim, Shephard and Chib \(1998\)](#), [Gordon, Salmond and Smith \(1993\)](#), [Pitt and Shephard \(1999\)](#), [Andrieu, Doucet and Holenstein \(2010\)](#), [Primiceri \(2005\)](#), [Morley, Nelson and Zivot \(2003\)](#), [Stock and Watson \(2007\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_06](#)
 - ▶ Kalman filter and smoother with exact diffuse initialisation, simulation smoothers, KSC Gibbs sampler, bootstrap and auxiliary particle filters, PMMH and UC-SV written out in `numpy`; checked against `statsmodels`
- Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Use
US GDP price deflator, real GDP (quarterly)	FRED (GDPDEF, GDPC1)	likelihood, MNZ, UC-SV
S&P 500 and BET, daily closes	EODHD	stochastic volatility
HICP of Romania and of the euro area (monthly index, 2025 = 100)	Eurostat prc_hicp_minr	TVP, UC-SV
Romanian real GDP, seasonally and calendar adjusted	Eurostat namq_10_gdp	output gap
US industrial production, payrolls, real income less transfers, real sales (monthly)	FRED (INDPRO, PAYEMS, W875RX1, CMRMTSPL)	DFM

- All sources are public and need no account or key; quarterly series to 2026Q2, monthly series to August or September 2026, daily data to 18 September 2026
- HICP: harmonised index of consumer prices; DFM: dynamic factor model

From Apollo navigation to trend inflation



Rudolf E. Kálmán, 2007

Photo: ETH-Bibliothek Zürich, Bildarchiv (2007); CC

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- 1960: [Kalman \(1960\)](#) gives the recursive linear filter; it guides the Apollo missions within a decade
- 1982–1997: EM for state space models [Shumway and Stoffer \(1982\)](#); structural time series [Harvey \(1990\)](#); exact diffuse filtering [Koopman \(1997\)](#)
- 1993–1999: particle filters [Gordon, Salmond and Smith \(1993\)](#), [Pitt and Shephard \(1999\)](#); Gibbs sampling of states [Carter and Kohn \(1994\)](#), [Frühwirth-Schnatter \(1994\)](#); stochastic volatility [Kim, Shephard and Chib \(1998\)](#)
- 2002–2010: the simulation smoother [Durbin and Koopman \(2002\)](#); TVP-VAR with stochastic volatility [Primiceri \(2005\)](#), [Cogley and Sargent \(2005\)](#); particle MCMC [Andrieu, Doucet and Holenstein \(2010\)](#)
- Today: central banks track trend inflation and the output gap with these tools [Stock and Watson \(2007\)](#), [Morley, Nelson and Zivot \(2003\)](#)

From TSA to this chapter

- Known ([TSA, Chapter 10](#)): the state space form, the Kalman filter for the local level by hand, smoothing, ML by the prediction error decomposition, the HP filter as a smoother, Hamilton (2018), Markov switching
- New here
 - ▶ the filter and smoother for any linear Gaussian model, with nonstationary states initialised exactly (not with a large variance)
 - ▶ inference when a variance may be zero; Bayesian sampling of whole state paths
 - ▶ nonlinear and non-Gaussian models: stochastic volatility, particle filters, particle MCMC
 - ▶ applications at research level: MNZ, UC-SV, TVP regressions, the ragged edge of a DFM
- Markov switching in depth: [Chapter 7](#); nowcasting with factors: [Chapter 5](#)

The general state space form (1/2)

- Two equations: the data are a noisy linear function of an unobserved state, and the state follows a first-order linear dynamic

$$y_t = Z_t \alpha_t + \varepsilon_t, \quad \alpha_{t+1} = c_t + T_t \alpha_t + R_t \eta_t, \quad t = 1, \dots, n$$

- Notation

- ▶ y_t ($p \times 1$): the observations; α_t ($m \times 1$): the state; n : the sample length
- ▶ Z_t ($p \times m$): the measurement matrix; T_t ($m \times m$): the transition matrix; c_t : a known intercept
- ▶ $\varepsilon_t \sim N(0, H_t)$: measurement noise; $\eta_t \sim N(0, Q_t)$: state shocks; R_t ($m \times r$) selects the states that receive shocks
- ▶ ε_t and η_s are independent of each other and over time
- Initial state $\alpha_1 \sim N(a_1, P_1)$
 - ▶ nonstationary or unknown fixed elements: $P_1 = \kappa P_\infty + P_*$, $\kappa \rightarrow \infty$; P_∞ marks the diffuse elements, P_* the covariance of the others

The general state space form (2/2)

- The system matrices Z_t, T_t, R_t, H_t, Q_t depend on parameters ψ ; time variation in Z_t gives regressions with time-varying coefficients
 - ▶ ARMA, UC, TVP regressions, DFM, VAR with missing data, ETS: all are special cases [Durbin and Koopman \(2012\)](#)
- Notation for the estimates
 - ▶ $Y_t = (y_1, \dots, y_t)$: the data up to t
 - ▶ predicted state: $a_t = \mathbb{E}(\alpha_t | Y_{t-1})$, $P_t = \text{Var}(\alpha_t | Y_{t-1})$; filtered: $a_{t|t}$, $P_{t|t}$ (after y_t)
 - ▶ smoothed: $\hat{\alpha}_t = \mathbb{E}(\alpha_t | Y_n)$, $V_t = \text{Var}(\alpha_t | Y_n)$ (all the data)

The Kalman filter from one lemma

- Lemma (conditioning of jointly normal vectors): if (x, y) is Gaussian, $\mathbb{E}(x|y) = \mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y)$,
 $\text{Var}(x|y) = \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}$
 - ▶ μ_x, μ_y : the means; $\Sigma_{xy} = \text{Cov}(x, y)$, $\Sigma_{yy} = \text{Var}(y)$: the covariance blocks
 - ▶ apply it to $x = \alpha_t$ and $y = y_t$, conditional on Y_{t-1} : $v_t = y_t - Z_t a_t$ is the new information (the prediction error)
- Prediction-error variance $\text{Var}(v_t|Y_{t-1}) = F_t = Z_t P_t Z_t' + H_t$; $\text{Cov}(\alpha_t, v_t|Y_{t-1}) = P_t Z_t'$; v_t is independent of Y_{t-1}
- Update: $a_{t|t} = a_t + P_t Z_t' F_t^{-1} v_t$, $P_{t|t} = P_t - P_t Z_t' F_t^{-1} Z_t P_t$
- Prediction: $a_{t+1} = c_t + T_t a_{t|t}$, $P_{t+1} = T_t P_{t|t} T_t' + R_t Q_t R_t'$
- Without normality the same recursions give the best *linear* predictor (minimum mean square error among linear functions of Y_t)

The filter in one pass

- ▣ $v_t = y_t - Z_t a_t$, $F_t = Z_t P_t Z_t' + H_t$, $K_t = T_t P_t Z_t' F_t^{-1}$ (Kalman gain), $L_t = T_t - K_t Z_t$
- ▣ $a_{t+1} = c_t + T_t a_t + K_t v_t$, $P_{t+1} = T_t P_t L_t' + R_t Q_t R_t'$
- ▣ Cost: $O(n(m^3 + p^3))$; with time-invariant matrices P_t converges to the steady state of the Riccati equation
 - ▶ P_t and K_t do not depend on y : they can be computed before the data arrive
- ▣ Missing y_t : set $Z_t = 0$ (or drop the missing rows): v_t and K_t vanish and the filter only predicts
 - ▶ mixed frequencies and the ragged edge of real-time data are therefore free ([Chapter 5](#))
- ▣ Multivariate y_t with diagonal H_t : process the p elements one at a time (univariate treatment [Koopman and Durbin \(2000\)](#)): no $p \times p$ inversion

Smoothing: backward recursions

- ▣ State smoother [Durbin and Koopman \(2012\)](#): $r_{t-1} = Z_t' F_t^{-1} v_t + L_t' r_t$, $N_{t-1} = Z_t' F_t^{-1} Z_t + L_t' N_t L_t$, $r_n = 0$, $N_n = 0$
- ▣ Smoothed state: $\hat{\alpha}_t = a_t + P_t r_{t-1}$, $V_t = P_t - P_t N_{t-1} P_t$
 - ▶ r_{t-1} : a weighted sum of the innovations from t on; $N_{t-1} = \text{Var}(r_{t-1})$
 - ▶ $\hat{\alpha}_t$ corrects a_t by everything learned from t on
- ▣ Disturbance smoother: $\hat{\varepsilon}_t = H_t(F_t^{-1} v_t - K_t' r_t)$, $\hat{\eta}_t = Q_t R_t' r_t$: auxiliary residuals for outliers and breaks, and the score of the likelihood
- ▣ Derivation from the same lemma, now conditioning on $v_t, \dots, v_n$, which are independent ([Appendix](#))

Diffuse initialisation (1/2)

- Random walks, trends and regression coefficients have no stationary distribution: $P_1 = \kappa P_\infty + P_*$, $\kappa \rightarrow \infty$
 - ▶ “big kappa” ($\kappa = 10^6$ – 10^7) is an approximation with two failures: the likelihood contains $-\frac{1}{2} \ln \kappa$ terms, and round-off errors grow with κ
- Exact diffuse filter [Koopman \(1997\)](#), [Ansley and Kohn \(1985\)](#), [de Jong \(1991\)](#)
 - ▶ expand $P_t = \kappa P_{\infty,t} + P_{*,t} + O(\kappa^{-1})$ and keep both parts while $P_{\infty,t} \neq 0$
 - ▶ $P_{\infty,t}$ reaches zero after d steps (d = number of diffuse elements, if they are identified)
 - ▶ then the ordinary filter takes over

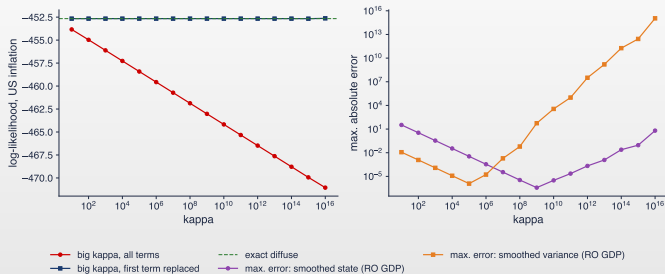
Diffuse initialisation (2/2)

□ The exact diffuse recursions

- ▶ $F_{\infty,t}, F_{*,t}$: the diffuse and finite parts of F_t ; $K_t^{(0)}, K_t^{(1)}$: the matching gains; $L_t^{(0)} = T_t - K_t^{(0)}Z_t, L_t^{(1)} = -K_t^{(1)}Z_t$
- ▶ univariate $y_t, F_{\infty,t} = Z_t P_{\infty,t} Z_t' > 0$: $K_t^{(0)} = T_t P_{\infty,t} Z_t' / F_{\infty,t}, K_t^{(1)} = T_t (P_{*,t} Z_t' - P_{\infty,t} Z_t' F_{*,t} / F_{\infty,t}) / F_{\infty,t}$
- ▶ $a_{t+1} = c_t + T_t a_t + K_t^{(0)} v_t; P_{\infty,t+1} = T_t P_{\infty,t} L_t^{(0)'}; P_{*,t+1} = T_t P_{\infty,t} L_t^{(1)'} + T_t P_{*,t} L_t^{(0)'} + R_t Q_t R_t'$

- The smoother has matching diffuse recursions for $r^{(0)}, r^{(1)}, N^{(0)}, N^{(1)}, N^{(2)}$ [Durbin and Koopman \(2012\)](#), Section 5.3; in the Quantlet both are checked against `statsmodels`

Exact diffuse against big kappa



- Left: local level for US GDP-deflator inflation at the ML variances; right: local linear trend for 100×log Romanian real GDP, $T = 106$; both against the exact diffuse filter

Interpreting the diffuse initialisation

- ▣ With all terms, the big-kappa log-likelihood falls by $\frac{1}{2} \ln 10$ per decade of κ : -460.72 at $\kappa = 10^7$, -468.78 at 10^{14} ; the exact value is -452.66
- ▣ Replacing the first term by its diffuse limit repairs the level (-452.66 at 10^7), but only because $d = 1$ is known; ML over ψ is unaffected, model comparison across different d is not
- ▣ Smoothed variances: error 0.000012 at $\kappa = 10^4$, 0.06 at 10^8 and 10^{15} at 10^{16} : no single κ is safe for both states and variances
- ▣ Practical rule: use the exact diffuse filter whenever the model has nonstationary or fixed unknown states

Recap: The general linear Gaussian model

- One lemma (Gaussian conditioning) gives the filter, the smoother and the disturbance smoother
- Missing data, mixed frequencies and multivariate observations are handled inside the same recursions
- Nonstationary states need the exact diffuse filter; big kappa breaks the likelihood level and the smoothed variances

The prediction error decomposition, diffuse case

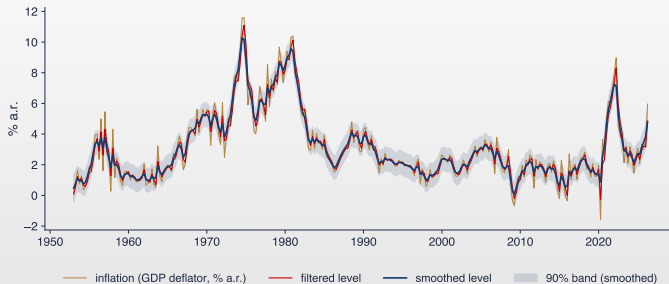
- $p(y_1, \dots, y_n) = \prod_t p(y_t | Y_{t-1})$, and $y_t | Y_{t-1} \sim N(Z_t a_t, F_t)$: the likelihood is a by-product of the filter
- Diffuse log-likelihood [Durbin and Koopman \(2012\)](#), eq. (7.4):

$$\ln L_d = -\frac{np}{2} \ln 2\pi - \frac{1}{2} \sum_{t=1}^d w_t - \frac{1}{2} \sum_{t=d+1}^n (\ln |F_t| + v_t' F_t^{-1} v_t)$$
- $w_t = \ln F_{\infty, t}$ if $F_{\infty, t} > 0$, otherwise $w_t = \ln F_{*, t} + v_t^2 / F_{*, t}$: the first d observations only pin down the diffuse states
 - ▶ equivalent to the marginal likelihood of [de Jong \(1991\)](#) and to treating the diffuse states as fixed and concentrating them out
- Scale: write $H_t = \sigma^2 H_t^*$, $Q_t = \sigma^2 Q_t^*$; then $F_t = \sigma^2 F_t^*$ and $\hat{\sigma}^2 = \frac{1}{p(n-d)} \sum_{t>d} v_t' F_t^{*-1} v_t$: one parameter fewer in the numerical search

Maximising the likelihood

- Parameterise variances as $\exp(\cdot)$, correlations as $\tanh(\cdot)$, AR coefficients through partial autocorrelations: unconstrained search, stationary models only
 - ▶ several starting values: UC likelihoods are often multimodal (MNZ below)
- Score in one filter-smoother pass: $\partial \ln L / \partial \psi = \frac{1}{2} \sum_t \text{tr} \{ (\hat{\varepsilon}_t \hat{\varepsilon}_t' + \text{Var}(\varepsilon_t | Y_n) - H_t) H_t^{-1} \frac{\partial H_t}{\partial \psi} H_t^{-1} \} + \dots$ [Durbin and Koopman \(2012\)](#), Section 7.3
- EM algorithm [Shumway and Stoffer \(1982\)](#): the E-step is the smoother, the M-step has closed forms for H and Q ; slow near the optimum but robust from poor starts (large DFM)
- Standard errors: inverse Hessian at $\hat{\psi}$ (numerical); valid only for interior points; report them on the transformed scale
- Identification: some UC models are observationally equivalent to ARIMA models with fewer parameters; check the reduced form before interpreting the components

Local level for US inflation: numpy against statsmodels



- GDP-deflator inflation, 1953Q1–2026Q2, $T = 294$; exact diffuse ML; the smoothed level with a 90% band and the one-sided (filtered) level

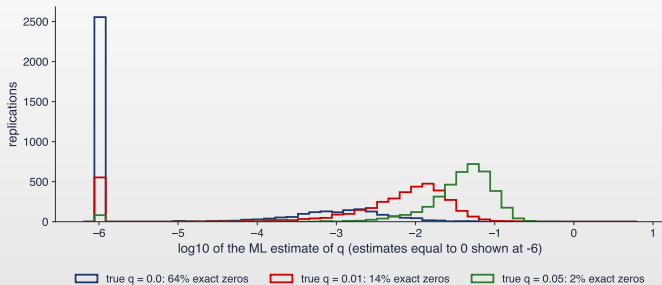
Interpreting the local level estimates

- ▣ numpy: $\hat{\sigma}_\varepsilon^2 = 0.517$, $\hat{\sigma}_\eta^2 = 0.453$, $\hat{q} = 0.877$; statsmodels: 0.517 and 0.453: the same optimum
- ▣ Log-likelihoods -452.66 and -451.74 : they differ by exactly $\frac{1}{2} \ln 2\pi = 0.919$, the constant that statsmodels omits for the $d = 1$ diffuse observation
- ▣ Steady-state gain 0.596: each new quarter moves the level estimate by 60% of the surprise; inflation is closer to a random walk than to white noise around a mean
- ▣ Last smoothed level 4.83% (s.d. 0.55): at the end of the sample the smoothed and filtered estimates coincide, and so does their uncertainty
- ▣ One constant variance for 1953–2026 is implausible (Great Inflation, Great Moderation): this motivates UC-SV in the last section

Inference at the boundary: the pile-up problem

- Local level with $q = \sigma_{\eta}^2 / \sigma_{\varepsilon}^2$: the ML estimate equals exactly zero with positive probability even when $q > 0$
Shephard and Harvey (1990)
 - ▶ the score at $q = 0$ is often negative because a smooth trend fits the data better than a slightly wandering one
- Consequences
 - ▶ the confidence interval computed from the Hessian is invalid
 - ▶ the LR test of $q = 0$ is not χ_1^2 , not even $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$: the model under the alternative is nonstationary
 - ▶ remedies: median-unbiased estimation of a coefficient variance Stock and Watson (1998); parametric bootstrap of the LR statistic (TVP section); Bayesian priors that keep q away from zero
- Same issue in TVP regressions (is the coefficient constant?) and in UC models (is the trend deterministic?)

How often is the estimated level variance exactly zero?



- 4 000 simulated local level series of length $T = 100$ for each true q ; ML of q on a fine grid with σ_ε^2 concentrated out

Interpreting the pile-up simulation

- True $q = 0$: 64% of the estimates are exactly zero; the rest spread over several orders of magnitude
- True $q = 0.01$: still 14% zeros; median estimate 0.0077; with $q = 0.05$: 2% zeros
- A reported $\hat{\sigma}_\eta^2 = 0$ is therefore not evidence of a constant level; a small positive value is compatible with a deterministic one
- Report the profile likelihood or the posterior of q , not a point estimate with a Hessian standard error

Recap: The exact likelihood

- The filter delivers the Gaussian likelihood; with diffuse states the first d terms are replaced by their diffuse limits
- Concentrate the scale, parameterise without constraints, use several starts, and check the numbers against a second implementation
- Variances near zero pile up at zero: standard errors and χ^2 tests fail at the boundary

Why sample the states?



Portrait traditionally identified as
Thomas Bayes (identification uncertain)

Portrait: unknown author; public domain;

[Wikimedia Commons](#)

- Target: $p(\alpha_1, \dots, \alpha_n, \psi | Y_n)$; the filter gives $p(\alpha_t | Y_t, \psi)$ for one t and one ψ only
- Gibbs sampler: alternate $\alpha_{1:n} \sim p(\alpha_{1:n} | \psi, Y_n)$ and $\psi \sim p(\psi | \alpha_{1:n}, Y_n)$; given the states, ψ is often a conjugate regression problem
- Uncertainty about ψ enters the bands of the states (ML bands ignore it)
- Functions of whole paths become easy: the date of a peak, the probability that a trend stays inside a target band
- Sampling the whole path in one block (not α_t one at a time) is what makes the sampler mix

Forward filtering, backward sampling

- Factorisation: $p(\alpha_{1:n}|Y_n) = p(\alpha_n|Y_n) \prod_{t=1}^{n-1} p(\alpha_t|\alpha_{t+1}, Y_t)$ (Markov property of the state)
- Forward: run the filter, store $a_{t|t}, P_{t|t}$; draw $\alpha_n \sim N(a_{n|n}, P_{n|n})$ [Carter and Kohn \(1994\)](#), [Frühwirth-Schnatter \(1994\)](#)
- Backward: $\alpha_t|\alpha_{t+1}, Y_t \sim N(m_t, S_t)$ by the lemma, with the smoothing gain $G_t = P_{t|t} T_t' P_{t+1}^{-1}$ (m_t, S_t : conditional mean and variance):
 - ▶ $m_t = a_{t|t} + G_t(\alpha_{t+1} - c_t - T_t a_{t|t}), \quad S_t = P_{t|t} - G_t T_t P_{t|t}$
- Issues: P_{t+1} is singular when $R_t Q_t R_t'$ is (lags in the state): use the rows with shocks only; n draws of dimension m
- Same output as the Gibbs step in the Bayesian VAR and DFM samplers of [Chapter 5](#)

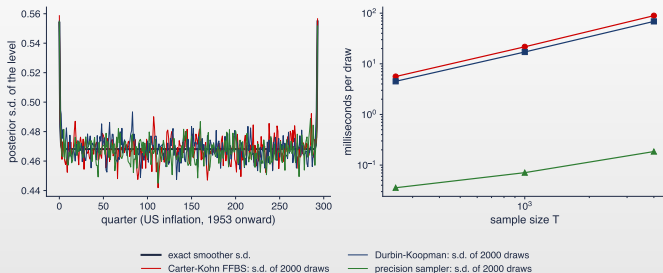
The Durbin–Koopman simulation smoother

- Idea [Durbin and Koopman \(2002\)](#): $\alpha - \hat{\alpha}(y)$ has the same conditional distribution for every y (Gaussian model): simulate it once from the model
- Algorithm (mean correction):
 - ▶ 1. draw $\alpha_{1:n}^+, y_{1:n}^+$ from the model (shocks and initial state); keep the missing pattern of y
 - ▶ 2. run one state smoother on $y - y^+$ (zero intercepts): $\hat{\alpha}(y - y^+) = \hat{\alpha}(y) - \hat{\alpha}(y^+)$
 - ▶ 3. $\tilde{\alpha} = \alpha^+ + \hat{\alpha}(y) - \hat{\alpha}(y^+)$ is a draw from $p(\alpha|Y_n)$
- Only the mean smoother is needed: no $P_{t|t}$ inversions, no singularity problems, diffuse states handled by the exact diffuse smoother (proof of exactness: [▶ Appendix](#))
- Precursor: the disturbance simulation smoother of [de Jong and Shephard \(1995\)](#); same idea for the disturbances ε_t, η_t

The precision sampler

- Stack the model [Chan and Jeliazkov \(2009\)](#): $y = X\alpha + \varepsilon$, $D\alpha = \eta$
 - ▶ D : a banded difference matrix (first differences for a random walk)
 - ▶ then $\alpha|y \sim N(\hat{\alpha}, K^{-1})$ with $K = D'\Sigma_{\eta}^{-1}D + X'\Sigma_{\varepsilon}^{-1}X$
- K is banded (tridiagonal for a random walk): one banded Cholesky factor $K = U'U$ costs $O(n)$
 - ▶ α , y : all states and observations stacked; X : block-diagonal of the Z_t ; Σ_{ε} , Σ_{η} : the stacked noise covariances; K : the posterior precision (inverse covariance)
 - ▶ $\hat{\alpha}$ solves $K\hat{\alpha} = X'\Sigma_{\varepsilon}^{-1}y$; a draw is $\hat{\alpha} + U^{-1}z$, $z \sim N(0, I)$
- Time-varying variances (stochastic volatility) only change the diagonal weights: ideal inside Gibbs samplers for UC-SV and SV
- Three samplers, one target: FFBS, DK and precision sampling draw from the same $p(\alpha|Y_n, \psi)$; they differ in cost and generality

Three samplers of the same posterior



- Local level for US inflation at the ML variances, proper prior $\mu_1 \sim N(y_1, 100)$; left: s.d. of 2000 draws against the exact smoother; right: time per draw for simulated samples

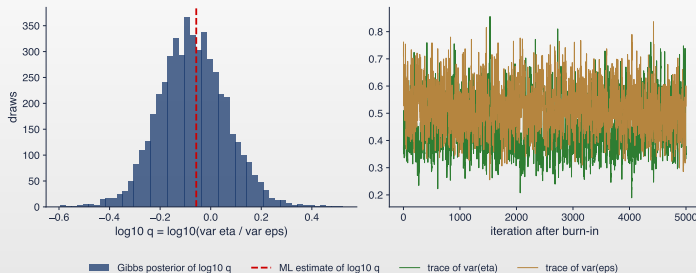
Interpreting the three samplers

- ▣ Ratios of draw s.d. to exact s.d. lie in $[0.94; 1.05]$ (FFBS), $[0.96; 1.05]$ (DK) and $[0.95; 1.04]$ (precision): Monte Carlo noise only
- ▣ Largest error of the draw means: 0.031, 0.027, 0.039, of the order of the Monte Carlo standard error
- ▣ For $T = 4000$: 89.2 ms (FFBS), 68.8 ms (DK), 0.19 ms (precision) per draw; all linear in T , the banded solver has no Python loop
- ▣ Choose DK for general models with diffuse states, precision sampling for random walks with time-varying variances

A Gibbs sampler for the local level model

- Model: $y_t = \mu_t + \varepsilon_t$, $\mu_{t+1} = \mu_t + \eta_t$; μ_t : the level; $IG(a_0, b_0)$: inverse gamma prior with shape a_0 and scale b_0 ; $q = \sigma_\eta^2 / \sigma_\varepsilon^2$
- Block 1: $\mu_{1:n} | \sigma_\varepsilon^2, \sigma_\eta^2, Y_n$ by the DK simulation smoother (exact diffuse μ_1)
- Block 2: $\sigma_\varepsilon^2 | \mu, Y_n \sim IG(a_0 + \frac{n}{2}, b_0 + \frac{1}{2} \sum_t (y_t - \mu_t)^2)$; $\sigma_\eta^2 | \mu \sim IG(a_0 + \frac{n-1}{2}, b_0 + \frac{1}{2} \sum_t (\Delta \mu_t)^2)$
- Prior $IG(2.5, 0.25)$ for both: proper, weak, and it rules out $\sigma_\eta^2 = 0$ exactly
 - the prior is not innocent near the boundary: a conjugate IG pushes small variances up; the non-centred parameterisation of [Frühwirth-Schnatter and Wagner \(2010\)](#) allows a normal prior on $\pm \sigma_\eta$ instead
- Convergence: trace plots, several chains, inefficiency factor $1 + 2 \sum_k \rho_k$ (Parzen weights): the number of draws worth one independent draw [Kim, Shephard and Chib \(1998\)](#)

Gibbs posterior of the signal-to-noise ratio



- US inflation, local level, 5 000 draws after burn-in; left: posterior of $\log_{10} q$ and the ML value; right: traces of the two variances

Interpreting the Gibbs output

- Posterior median of q : 0.84, 90% credible interval [0.52; 1.42]; ML: 0.88
- Posterior means 0.515 and 0.441 against ML 0.517 and 0.453: with $n = 294$ the prior barely matters away from the boundary
- Inefficiency factors 9.9 and 11.3: the variance draws inherit the autocorrelation of the state draws
- The interval for q excludes zero: the trend of US inflation moves, but the upper end of the interval is 2.7 times the lower end

Recap: Bayesian state space

- Gibbs: states as one block given the parameters, parameters given the states
- FFBS, the DK simulation smoother and the precision sampler draw from the same distribution
- Priors regularise boundary problems but must be reported and varied

The stochastic volatility model



Neil Shephard, 2004

Photo: Renate Schmid (2004), MFO; CC

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- $y_t = \exp(h_t/2) \epsilon_t$, $h_{t+1} = \mu + \phi(h_t - \mu) + \sigma_\eta \eta_t$, $\epsilon_t, \eta_t \sim \text{i.i.d. } N(0, 1)$ Taylor (1994)
- h_t : log-variance, a latent AR(1) state; ϕ : persistence; σ_η : volatility of volatility; $\exp(\mu/2)$: typical volatility
- A state space model with a nonlinear measurement equation: the Kalman filter does not apply directly
- Continuous-time counterpart: the log-variance as an Ornstein–Uhlenbeck process (option pricing, realised measures in Chapter 8)

Stochastic volatility is not GARCH (1/2)

- GARCH [Bollerslev \(1986\)](#): $\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2$, $\omega > 0$, $\alpha, \beta \geq 0$
 - ▶ σ_t^2 is known at $t - 1$ (a function of past returns): one source of randomness
 - ▶ the likelihood is a product of known densities: one pass, exact ML
- SV: h_t has its own shock η_t ; even with all past returns, today's variance is uncertain
 - ▶ $p(y_{1:n}|\psi) = \int \prod_t N(y_t; 0, e^{h_t}) p(h_{1:n}|\psi) dh_{1:n}$: an n -dimensional integral without closed form
- Comparison on the likelihood scale is possible only after the integral is computed: particle filter (next section) or importance sampling [Kim, Shephard and Chib \(1998\)](#)

Stochastic volatility is not GARCH (2/2)

- Moments, with $\sigma_h^2 = \sigma_\eta^2 / (1 - \phi^2)$ the unconditional variance of h_t
 - ▶ kurtosis of y_t : $3 \exp(\sigma_h^2) > 3$ even with Gaussian ϵ_t : fat tails come from the random variance
 - ▶ $\text{Corr}(y_t^2, y_{t-k}^2) = (e^{\sigma_h^2 \phi^k} - 1) / (3e^{\sigma_h^2} - 1)$: volatility clustering that decays with ϕ^k
- Leverage enters as $\text{Corr}(\epsilon_t, \eta_t) = \rho < 0$ [Omori, Chib, Shephard and Nakajima \(2007\)](#); GARCH needs an asymmetric term for the same effect

A linear state space form: log-squared returns

- $y_t^* = \ln y_t^2 = h_t + \xi_t$, $\xi_t = \ln \epsilon_t^2$: linear in h_t , but ξ_t follows a $\ln \chi_1^2$ law with mean -1.2704 and variance $\pi^2/2 \approx 4.93$
- QML [Harvey, Ruiz and Shephard \(1994\)](#): run the Kalman filter as if ξ_t were $N(-1.2704, \pi^2/2)$; consistent, but inefficient
 - ▶ y_t^* is ARMA(1,1): $\text{Corr}(y_t^*, y_{t-k}^*) = \phi^k \sigma_h^2 / (\sigma_h^2 + \pi^2/2)$, small because the noise dominates the signal
- Zero returns: $\ln 0 = -\infty$; use $y_t^* = \ln(y_t^2 + c)$ with a small offset c (here $c = 10^{-3} \text{Var}(y)$, scaled to the series)
- Multivariate extension of the same idea: factor SV models [Harvey, Ruiz and Shephard \(1994\)](#)

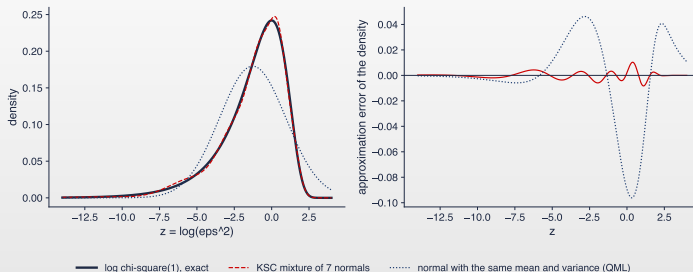
The KSC mixture: a conditionally Gaussian model

- Kim, Shephard and Chib (1998): approximate $\ln \chi_1^2$ by a mixture of seven normals; given the component s_t , the model is linear and Gaussian
- $y_t^* = h_t + m_{s_t} - 1.2704 + \sqrt{v_{s_t}} u_t$, $u_t \sim N(0, 1)$, $P(s_t = i) = p_i$; $s_t \in \{1, \dots, 7\}$: the component; p_i , m_i , v_i : its probability, mean and variance (table)
- Mixture mean -1.2704 and variance 4.935 , against -1.2704 and 4.935 for the exact law
- The residual approximation error can be removed by reweighting the draws (importance weights) Kim, Shephard and Chib (1998); finer 10-component mixtures Omori, Chib, Shephard and Nakajima (2007)

i	p_i	m_i	v_i
1	0.00730	-10.12999	5.79596
2	0.10556	-3.97281	2.61369
3	0.00002	-8.56686	5.17950
4	0.04395	2.77786	0.16735
5	0.34001	0.61942	0.64009
6	0.24566	1.79518	0.34023
7	0.25750	-1.08819	1.26261

- Source: Kim, Shephard and Chib (1998), Table 4

The log chi-square law and its approximations



- Left: density of $\ln \epsilon^2$ (exact), the KSC mixture and the normal with the same two moments used by QML; right: the two approximation errors

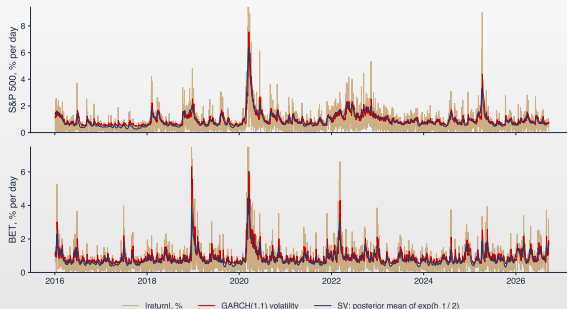
Interpreting the mixture approximation

- ▣ The law is strongly skewed to the left: small returns produce very negative $\ln y_t^2$; the normal approximation misses both the skew and the peak
- ▣ Largest density error: 0.010 for the mixture, 0.096 for the normal (QML)
- ▣ QML treats a skewed noise as symmetric: the filter overreacts to near-zero returns; the mixture fixes this at the cost of one extra Gibbs block

Gibbs sampling of the SV model

- Block 1: $s_t|y_t^*, h_t$ independently, $P(s_t = i) \propto p_i N(y_t^* - h_t; m_i - 1.2704, v_i)$
- Block 2: $h_{1:n}|s, \psi, y^*$ in one draw: a linear Gaussian model with known time-varying measurement variances v_{s_t} (precision sampler or DK)
- Block 3: $\sigma_\eta^2|h, \mu, \phi$ inverse gamma; ϕ by Metropolis–Hastings with the AR(1) regression as proposal; μ normal
- Priors [Kim, Shephard and Chib \(1998\)](#): $(\phi + 1)/2 \sim \text{Beta}(20, 1.5)$, $\sigma_\eta^2 \sim \text{IG}(2.5, 0.025)$, $\mu \sim N(0, 10)$
- Single-move samplers (one h_t at a time, [Jacquier, Polson and Rossi \(1994\)](#)) mix very slowly because consecutive h_t are highly correlated: sample the path as a block

Stochastic volatility of the S&P 500 and the BET



- Daily log returns in %, demeaned, 2016–2026 ($T = 2692$ and 2675); KSC Gibbs sampler, 20 000 draws; GARCH(1,1) by ML on the same data

Interpreting the SV estimates

Index	ϕ	σ_η	μ	ineff. ϕ, σ_η	GARCH $\alpha + \beta$	corr. vol.
S&P 500	0.963	0.289	-0.465	49 / 98	0.967	0.91
BET	0.914	0.393	-0.624	58 / 87	0.952	0.84

- 90% intervals for ϕ : [0.950; 0.975] (S&P 500), [0.888; 0.938] (BET): very persistent log-variance, as in [Kim, Shephard and Chib \(1998\)](#)
- The two volatility paths are close (correlation in the last column), but SV reacts less to a single large return than GARCH, whose variance jumps by αy_{t-1}^2
- Large inefficiency factors for σ_η are typical: σ_η and the path $h_{1:n}$ are strongly dependent; long chains or interweaving are needed
- Kurtosis of returns: 19.5 (S&P 500) and 20.8 (BET); with Gaussian ϵ_t , SV explains only part of it (SV- t errors are the usual extension)

Recap: Stochastic volatility

- SV adds a shock to the variance: its likelihood is an integral over the volatility path
- Log-squared returns make the model linear; the KSC mixture makes it conditionally Gaussian, so the whole path can be drawn in one block
- On daily index returns SV and GARCH give similar volatility paths but different reactions to single large returns

The Bayes filter

- General model: $y_t \sim g(y_t|x_t, \psi)$, $x_t \sim f(x_t|x_{t-1}, \psi)$; g : measurement density, f : transition density, x_t : the state; target $p(x_t|Y_t)$ and $p(y_t|Y_{t-1})$
- Prediction: $p(x_t|Y_{t-1}) = \int f(x_t|x_{t-1}) p(x_{t-1}|Y_{t-1}) dx_{t-1}$
- Update: $p(x_t|Y_t) = g(y_t|x_t) p(x_t|Y_{t-1})/p(y_t|Y_{t-1})$, $p(y_t|Y_{t-1}) = \int g(y_t|x_t) p(x_t|Y_{t-1}) dx_t$
- Closed form only in two cases: linear Gaussian (Kalman) and finite state spaces (the Hamilton filter or Markov switching, [Chapter 7](#))
 - ▶ otherwise: approximate the model (EKF), approximate the densities by points (UKF), or by weighted samples (particle filters)

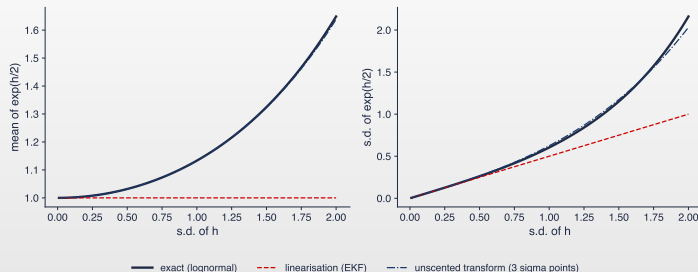
The extended Kalman filter

- Nonlinear model $y_t = Z(x_t) + \varepsilon_t$, $x_{t+1} = T(x_t) + R\eta_t$: linearise around the current estimates, $\dot{Z}_t = \partial Z / \partial x|_{a_t}$, $\dot{T}_t = \partial T / \partial x|_{a_t|_t}$
- Run the Kalman recursions with $\dot{Z}_t$, $\dot{T}_t$ for the variances and the exact functions for the means [Durbin and Koopman \(2012\)](#), Section 10.2
- A failure that teaches: in SV, $\mathbb{E}(y_t|h_t) = 0$ for every h_t , so $\partial \mathbb{E}(y_t|h_t) / \partial h_t = 0$: the EKF gain is zero and the filter never learns h_t
 - the information about h_t is in y_t^2 , a second moment; linearisation keeps only first moments
- Rule: transform the model first ($\ln y_t^2$), or use a method that propagates the whole distribution

The unscented Kalman filter

- ▣ Unscented transform [Julier and Uhlmann \(2004\)](#): $N(m, P)$ in n dimensions is represented by $2n + 1$ sigma points
 - ▶ $\mathcal{X}_0 = m$, $\mathcal{X}_{\pm i} = m \pm (\sqrt{(n + \lambda)P})_i$; $(\sqrt{A})_i$: the i -th column of a matrix square root
 - ▶ weights $w_0 = \lambda/(n + \lambda)$, $w_{\pm i} = 1/(2(n + \lambda))$; λ : a spread parameter
 - ▶ push each point through g and take weighted means and covariances
- ▣ Exact for the mean to second order of the Taylor expansion of g (the EKF: first order); no Jacobians needed
- ▣ UKF: apply the transform in the prediction and in the update step; cost similar to the EKF
- ▣ Still Gaussian at each step: multimodal or heavy-tailed filtering densities need particles

Linearisation against the unscented transform

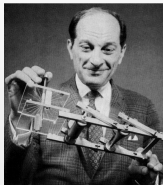


- Volatility from log-variance: mean and s.d. of $\exp(h/2)$ for $h \sim N(0, s^2)$; exact lognormal moments, first-order linearisation (EKF) and the unscented transform with three sigma points

Interpreting the unscented transform

- At $s = 1$: exact mean 1.131, linearisation 1.000, unscented 1.131; exact s.d. 0.598, linearisation 0.496, unscented 0.618
- Linearisation ignores the convexity of $\exp$: it underestimates expected volatility, more so when h is uncertain
- Three sigma points capture the convexity almost exactly for the mean; the variance is off for large s
- Same bias in practice: an EKF forecast of volatility or of an option price is too low when the state is uncertain

Particle filters



Stanisław Ulam with the FERMIAC, a
Monte Carlo device

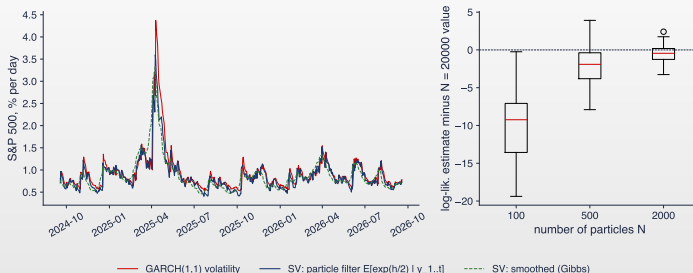
Photo: Los Alamos National Laboratory
(LA-UR-00-2532); public domain; Wikimedia
Commons

- Represent $p(x_t|Y_t)$ by particles $x_t^{(i)}$ with weights $W_t^{(i)}$, $i = 1, \dots, N$; Monte Carlo goes back to Ulam and Metropolis (1949)
- Bootstrap filter Gordon, Salmond and Smith (1993), Kitagawa (1996)
 - ▶ propagate: $x_t^{(i)} \sim f(\cdot|x_{t-1}^{(i)})$
 - ▶ weight: $w_t^{(i)} = g(y_t|x_t^{(i)})$
 - ▶ resample N particles with probabilities $W_t^{(i)} \propto w_t^{(i)}$
- Without resampling the weights degenerate (one particle takes all the weight); effective sample size $ESS_t = 1 / \sum_i (W_t^{(i)})^2$
- Systematic resampling: one uniform U , points $(U + i - 1)/N$ on the cumulative weights: lower variance than multinomial

The particle likelihood

- $\hat{p}(y_t|Y_{t-1}) = \frac{1}{N} \sum_i w_t^{(i)}$ and $\hat{L}(\psi) = \prod_t \hat{p}(y_t|Y_{t-1})$
- $\mathbb{E} \hat{L}(\psi) = L(\psi)$ exactly, for any $N \geq 1$ Del Moral (2004) (▶ [Appendix](#))
 - ▶ $\hat{L}$ is unbiased on the level scale
 - ▶ not on the log scale: if $\ln \hat{L} \approx N(\ln L - \tau^2/2, \tau^2)$, then $\mathbb{E} \ln \hat{L} = \ln L - \tau^2/2$; $\tau^2 = \text{Var} \ln \hat{L}$
- $\text{Var} \ln \hat{L}$ grows roughly linearly in n and falls like $1/N$: double the sample, double the particles
- $\hat{L}(\psi)$ is not smooth in ψ (resampling is discontinuous): do not maximise it by gradient methods; use it inside MCMC

Particle filter for the S&P 500 SV model



- Left: filtered volatility $E[\exp(h_t/2) | Y_t]$ (bootstrap filter, $N = 20\,000$, Gibbs posterior mean of ψ) against GARCH(1,1) and the smoothed SV path, last two years; right: log-likelihood estimates, 100 runs per N

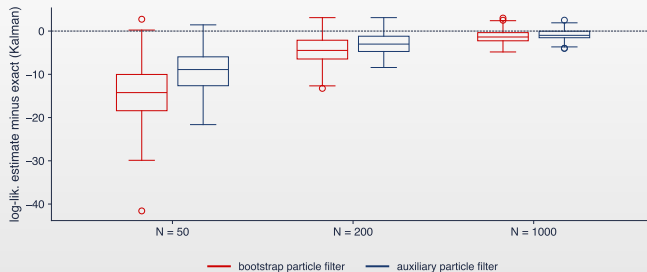
Interpreting the particle filter

- Monte Carlo s.d. of $\ln \hat{L}$: 4.19 ($N = 100$), 2.38 ($N = 500$), 1.14 ($N = 2000$); small N also biases $\ln \hat{L}$ downwards
- SV log-likelihood -3378.8 against the GARCH(1,1) maximum -3467.3 : a difference of 88.5 with the same number of parameters
- The comparison favours SV even at the posterior mean (not the maximum) of its likelihood; [Kim, Shephard and Chib \(1998\)](#) compare SV with ARCH models on the same likelihood scale
- The filtered SV volatility is a one-sided estimate, like GARCH: correlation 0.96; the smoothed path anticipates turning points because it uses future data
- Effective sample size over the sample: minimum 150, median 18719 of 20 000 particles

The auxiliary particle filter

- ▣ Bootstrap weakness: particles are propagated blind to y_t ; an outlier leaves few particles with weight
- ▣ APF Pitt and Shephard (1999): look ahead before propagating
 - ▶ first stage: resample $x_{t-1}^{(i)}$ with weights $\propto W_{t-1}^{(i)} g(y_t | \mu_t^{(i)})$, $\mu_t^{(i)} = \mathbb{E}(x_t | x_{t-1}^{(i)})$
 - ▶ second stage: propagate, then weight by $g(y_t | x_t^{(i)}) / g(y_t | \mu_t^{(k_i)})$ to correct the look-ahead
 - ▶ k_i : the index of the particle selected in the first stage
- ▣ Fully adapted case: first-stage weights $p(y_t | x_{t-1})$ and proposal $p(x_t | x_{t-1}, y_t)$: second-stage weights are equal (locally optimal)
- ▣ The likelihood estimate stays unbiased: $\hat{p}(y_t | Y_{t-1}) = (\sum_i W_{t-1}^{(i)} g(y_t | \mu_t^{(i)})) \cdot \frac{1}{N} \sum_i \omega_t^{(i)}$; $\omega_t^{(i)}$: the second-stage weights

Particle likelihoods against the exact Kalman likelihood



- Local level model for US inflation at the ML variances: error of $\ln \hat{L}$ (given y_1) for the bootstrap and the auxiliary filter, 200 runs per N

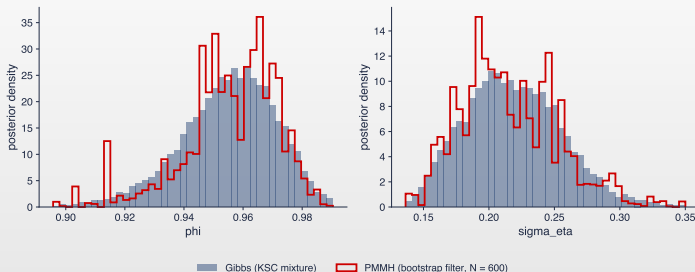
Interpreting the particle check

- Mean error of $\ln \hat{L}$: $-14.37, -4.48, -1.30$ (bootstrap, $N = 50, 200, 1000$) and $-9.37, -2.87, -0.88$ (auxiliary)
- Standard deviations: 1.46 (bootstrap) and 1.17 (auxiliary) at $N = 1000$: the look-ahead buys precision because the signal-to-noise ratio is high
- The negative mean is the Jensen bias of a log of an unbiased estimator: it shrinks like $\text{Var}(\ln \hat{L})/2$
- Always test a particle filter on a model with a known likelihood before using it where none is available

Particle MCMC

- Pseudo-marginal idea [Andrieu and Roberts \(2009\)](#): run Metropolis–Hastings with an unbiased, positive estimate $\hat{L}(\psi)$ in place of $L(\psi)$; the chain still targets the exact posterior $p(\psi|Y_n)$
- PMMH [Andrieu, Doucet and Holenstein \(2010\)](#)
 - ▶ propose ψ' from a proposal density $q(\cdot|\psi)$; run a particle filter for $\hat{L}(\psi')$
 - ▶ accept with probability $\min\{1, \hat{L}(\psi')p(\psi')q(\psi|\psi')/[\hat{L}(\psi)p(\psi)q(\psi'|\psi)]\}$; $p(\cdot)$: the prior
 - ▶ keep $\hat{L}$ of the current point (do not recompute it)
- Choosing N [Pitt, Silva, Giordani and Kohn \(2012\)](#), [Doucet, Pitt, Deligiannidis and Kohn \(2015\)](#)
 - ▶ too few particles make the chain stick; too many waste time
 - ▶ target: $\text{Var}(\ln \hat{L}) \approx 1\text{--}2$ at the posterior mean
 - ▶ particle Gibbs (conditional SMC) samples the state path too; useful when no conditionally Gaussian structure exists
- Works for any model you can simulate and whose measurement density you can evaluate: SV with jumps, DSGE models, nonlinear UC models

PMMH against the KSC Gibbs sampler



- Last 1000 S&P 500 returns; PMMH with a bootstrap filter of $N = 600$ particles, 10 000 iterations, random walk on $(\mu, \tanh \phi, \ln \sigma_{\eta})$; Gibbs with the KSC mixture

Interpreting the PMMH run

- Posterior means of ϕ : 0.955 (Gibbs) and 0.955 (PMMH); of σ_η : 0.219 and 0.216; posterior s.d. 0.037 and 0.039
- Two exact algorithms, one posterior: the KSC mixture approximation has no visible effect here
- $N = 600$ gives s.d. of $\ln \hat{L}$ equal to 1.19 at the posterior mean, inside the recommended range; acceptance rate 13%
- Cost: 3.6 minutes for PMMH against seconds for Gibbs: use PMMH when there is no conditionally Gaussian representation

Recap: Nonlinear filtering

- EKF linearises the model, UKF propagates sigma points, particle filters propagate weighted samples
- The particle likelihood is unbiased in levels; its log variance decides the number of particles
- PMMH turns any simulable model into an exact Bayesian procedure, at a computational price

A TVP regression in state space form

- $y_t = c_t + b_t x_t + D_t' \gamma + \varepsilon_t, \quad c_{t+1} = c_t + \eta_{c,t}, \quad b_{t+1} = b_t + \eta_{b,t}$
 - ▶ c_t, b_t : intercept and slope, random walks with variances σ_c^2, σ_b^2 ; D_t : dummies with fixed γ
 - ▶ state $\alpha_t = (c_t, b_t, \gamma')'$; $Z_t = (1, x_t, D_t')$
- Fixed coefficients (γ , here quarterly seasonal dummies) are states with zero variance and a diffuse start: $d =$ number of states = 5
 - ▶ with $\sigma_c^2 = \sigma_b^2 = 0$ the smoother returns the OLS estimates (recursive least squares)
- Question: has the co-movement of Romanian inflation with euro-area inflation changed since inflation targeting began (2005)?
- y_t : Romanian HICP inflation, quarterly, % a.r., not seasonally adjusted; x_t : euro-area HICP inflation net of its seasonal pattern; 2005Q4–2026Q2, $T = 83$
- Test of a constant slope: $H_0: \sigma_b^2 = 0$ on the boundary; parametric bootstrap of the LR statistic from the restricted model (199 samples)

Romanian on euro-area inflation: a time-varying slope?



- Top: the two inflation rates; bottom: smoothed b_t with a 90% band (exact diffuse ML) and the slope of a rolling 12-quarter regression with the same dummies

Interpreting the TVP regression

- ▣ ML: $\hat{\sigma}_c^2 = 0.332$, $\hat{\sigma}_b^2 = 0.0011$, $\hat{\sigma}_\varepsilon^2 = 7.21$; numpy and statsmodels log-likelihoods -209.93 and -209.93
- ▣ b_t drifts from 0.76 (s.e. 0.23) in 2008Q3 to 0.89 (s.e. 0.17) in 2022Q4: a slope close to one, but no significant change
- ▣ LR = 0.066; bootstrap $p = 0.265$ (95% quantile 1.91; 66% of bootstrap statistics are exactly zero: pile-up again); the $\frac{1}{2}\chi_1^2$ rule gives $p = 0.399$
- ▣ The rolling regression swings between negative values and 1.5: that is sampling noise of 12-quarter windows, not time variation; the state space model averages it out with an explicit variance

TVP-VAR with stochastic volatility



Thomas J. Sargent, 2011

Photo: Holger Motzkau (2011); CC BY-SA 3.0; Wikimedia Commons

- Cogley and Sargent (2001), Cogley and Sargent (2005), Primiceri (2005):
$$y_t = X_t' \beta_t + A_t^{-1} \Sigma_t \epsilon_t$$
 - ▶ X_t : the lags; β_t : the coefficients; $\epsilon_t \sim N(0, I)$
 - ▶ A_t : lower triangular with free elements a_t ; $\Sigma_t = \text{diag}(\sigma_{j,t})$
 - ▶ β_t , a_t and $\ln \sigma_{j,t}$ follow random walks
- Question: did US monetary policy change (drifting coefficients) or did the shocks shrink (falling volatilities) after 1980?
- Answer of Primiceri (2005): the systematic response of policy did change, but these changes explain little of the 1970s; the variance of non-policy shocks matters more
- Without stochastic volatility, falling shock variances are wrongly attributed to changing coefficients

Estimating a TVP-VAR-SV

- Gibbs blocks, each one of the samplers of this chapter:
 - ▶ $\beta_{1:n} \mid \text{rest}$: simulation smoother of a linear Gaussian model with known time-varying covariances
 - ▶ $a_{1:n} \mid \text{rest}$: equation by equation, the same smoother
 - ▶ $\ln \sigma_{j,1:n} \mid \text{rest}$: the KSC mixture with its indicators
 - ▶ covariances of the random walks: inverse Wishart
- The order of the Gibbs steps matters: the original algorithm drew the mixture indicators at the wrong point of the cycle; [Del Negro and Primiceri \(2015\)](#) give the corrected order, and the substantive results change little
- Priors from a training sample at the start of the data, as in [Primiceri \(2005\)](#), with small scales for the random-walk covariances: the prior decides how much variation is allowed
- Overparameterisation: with k coefficients per equation, shrink the variances, or select which coefficients vary (non-centred parameterisation, [Frühwirth-Schnatter and Wagner \(2010\)](#))

Recap: Time-varying parameters

- A TVP regression is a state space model with Z_t built from the regressors; fixed coefficients are diffuse states without shocks
- Rolling windows exaggerate time variation; test it at the boundary with a bootstrap
- TVP-VAR-SV combines the simulation smoother and the KSC mixture in one Gibbs sampler

The DFM as a state space model (1/2)

- $x_{it} = \lambda_i' f_t + e_{it}, \quad f_t = \Phi f_{t-1} + u_t, \quad e_{it} = \rho_i e_{i,t-1} + \nu_{it}$
 - ▶ f_t : the factors, a VAR(1) with matrix Φ ; λ_i : the loadings of series i , stacked in Λ
 - ▶ e_{it} : idiosyncratic AR(1) errors with coefficients ρ_i ; u_t, ν_{it} : shocks
 - ▶ state $(f_t', f_{t-1}', e_t')'$; measurement matrix $Z = (\Lambda, 0, I)$
- Mixed frequencies [Mariano and Murasawa \(2003\)](#)
 - ▶ a quarterly growth rate is a weighted sum of monthly states
 - ▶ missing months are simply skipped by the filter

The DFM as a state space model (2/2)

▣ Estimation

- ▶ two steps [Doz, Giannone and Reichlin \(2011\)](#): principal components give Λ , Φ , Σ_e ; then the Kalman smoother re-estimates f_t on the full panel; consistent as $N, T \rightarrow \infty$
 - ▶ ML by EM [Shumway and Stoffer \(1982\)](#), [Bańbura and Modugno \(2014\)](#): the E-step is the smoother, arbitrary missing patterns are allowed
 - ▶ large N : collapse the N observations into an r -dimensional vector before filtering [Jungbacker and Koopman \(2015\)](#); the cost no longer grows with N^3
- ▣ Nowcasting [Giannone, Reichlin and Small \(2008\)](#): the ragged edge, news decomposition and real-time evaluation are in [Chapter 5](#); this section shows the state space engine

A one-factor model of the US coincident indicators



- Monthly growth of industrial production, payrolls, real income less transfers and real sales (FRED), standardised, January 1990–September 2026; two-step estimator, univariate treatment of the observation vector; last 36 months

ATS_ch6_dfm_ragged_edge

Interpreting the ragged edge

- ▣ Loadings 0.90, 0.89, 0.68, 0.84; factor AR coefficient 0.12: monthly growth rates have little persistence, so the factor is mostly a weighted average of the month
- ▣ The last complete month is July 2026: the factor s.d. is 0.28; in September 2026, with one series of four available, it rises to 0.46
- ▣ The filter weighs each series by its signal-to-noise ratio; no series is dropped, no gap is filled by hand
- ▣ Forecasting the factor beyond the data is the same recursion with all observations missing

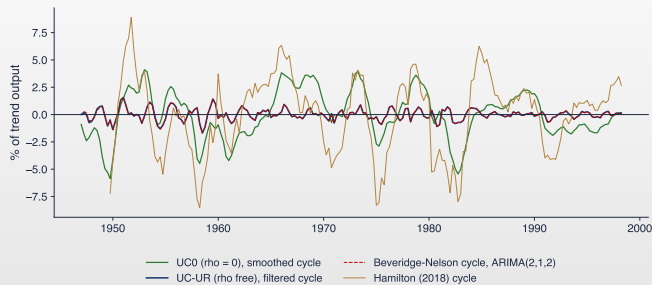
Three definitions of the cycle

- Beveridge–Nelson [Beveridge and Nelson \(1981\)](#): trend = long-horizon forecast net of drift
 - ▶ $\tau_t^{BN} = y_t + \sum_{j \geq 1} \mathbb{E}_t(\Delta y_{t+j} - \mu)$; cycle $c_t^{BN} = y_t - \tau_t^{BN}$
 - ▶ $\mathbb{E}_t$: expectation given data to t ; μ : mean growth; the trend is where y is expected to settle once the predictable growth has died out
 - ▶ for an AR(1) in growth: $c_t^{BN} = -\frac{\phi}{1-\phi}(\Delta y_t - \mu)$; one-sided, depends on the ARMA chosen; reliable versions: [Kamber, Morley and Wong \(2018\)](#)
- Unobserved components [Clark \(1987\)](#), [Watson \(1986\)](#), [Harvey and Jaeger \(1993\)](#): $y_t = \tau_t + c_t$ with a random-walk (or smooth) trend and an AR(2) cycle; two-sided after smoothing
- Hamilton [Hamilton \(2018\)](#): c_{t+h} = residual of y_{t+h} on $(1, y_t, \dots, y_{t-3})$, $h = 8$ quarters; model-free and one-sided ([TSA, Chapter 10](#))
- For US GDP the BN cycle is small and noisy, the UC cycle large and smooth: the same data, two stories. Why?

Unobserved components with correlated shocks

- Morley, Nelson and Zivot (2003), the UC-UR model
 - ▶ $y_t = \tau_t + c_t$, $\tau_t = \mu + \tau_{t-1} + \eta_t$, $c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + \varepsilon_t$, $\text{Corr}(\eta_t, \varepsilon_t) = \rho$
 - ▶ τ_t : the stochastic trend with drift μ and shocks η_t (s.d. σ_η); c_t : an AR(2) cycle with shocks ε_t (s.d. σ_ε); ρ : the correlation of the two shocks
- Clark's UC0 imposes $\rho = 0$; with an AR(2) cycle, ρ is identified: the reduced form is an ARIMA(2,1,2) with as many parameters as UC-UR
 - ▶ with an AR(1) cycle, ρ is not identified
- Result: if ρ is free, the filtered UC cycle equals the BN cycle of the ARIMA(2,1,2): the difference between BN and UC was the restriction $\rho = 0$
- State (τ_t, c_t, c_{t-1}) : τ diffuse, the cycle block stationary (mixed initialisation); case study below: their sample, 1947Q1–1998Q2, today's data vintage

Replicating Morley, Nelson and Zivot (2003)



- 100×log US real GDP (FRED GDPC1), $T = 206$; UC0 and UC-UR by exact diffuse ML; BN cycle from an ARIMA(2,1,2); Hamilton (2018) cycle for comparison

Interpreting the MNZ replication

Model	μ	ϕ_1	ϕ_2	σ_η	σ_ε	ρ	$\ln L$
UC0	0.858	1.501	-0.571	0.612	0.665	0	-280.81
UC-UR	0.859	1.334	-0.738	1.185	0.669	-0.927	-279.35

- $\hat{\rho} = -0.927$: trend and cycle shocks are almost perfectly negatively correlated; trend shocks are larger than cycle shocks ($\hat{\sigma}_\eta > \hat{\sigma}_\varepsilon$)
- Correlation of the UC-UR filtered cycle with the BN cycle (ARIMA AR roots 1.334, -0.738 as in UC-UR): 0.999; s.d. of the cycles: UC0 2.20, UC-UR 0.52, BN 0.52, Hamilton 3.50
- LR test of $\rho = 0$: 2.92, $p = 0.088$ (χ_1^2); the cycle of UC0 is large because it is assumed orthogonal to the trend, not because the data demand it
- Our numbers use today's GDP vintage; the paper's estimates differ in the decimals, not in the conclusion

Romania: the cycle is the trend



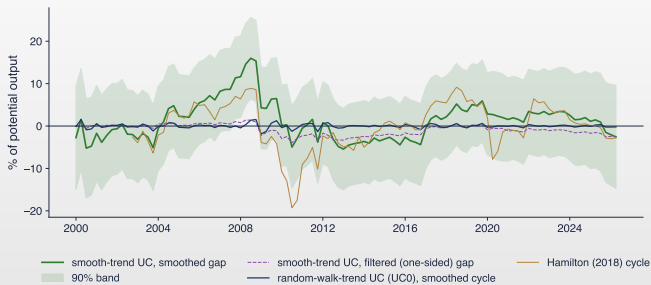
The BNR palace, Bucharest

Photo: Ștefan Jurcă (2015); CC BY 2.0;

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- Emerging economies have volatile trend shocks [Aguilar and Gopinath \(2007\)](#): a random-walk trend absorbs most of the movement in GDP
- Romanian UC0, 2000Q1–2026Q2 (2020Q2–Q4 treated as missing): $\hat{\sigma}_\eta = 1.897$, $\hat{\sigma}_\varepsilon = 0.950$, $\hat{\phi}_1 = 0.280$, $\hat{\phi}_2 = -0.261$: a short, weak cycle
- UC-UR: $\hat{\rho} = 1.00$ at the boundary, LR = 2.92: in 106 quarters the correlation is poorly determined
- Alternative: a smooth trend (integrated random walk, the HP model [Harvey and Jaeger \(1993\)](#)) with an AR(2) cycle: $\hat{\sigma}_\zeta^2 = 0.0046$, $\hat{\sigma}_\varepsilon^2 = 4.95$, $\hat{\phi} = (0.87, 0.09)$

The Romanian output gap



- 100×log real GDP, Eurostat, seasonally and calendar adjusted; smooth-trend UC (smoothed, with 90% band, and filtered), random-walk-trend UC0, Hamilton (2018) on the full series

Interpreting the Romanian output gap

- Smooth-trend gap: 16.0% in 2008Q3 (s.d. 5.9), -5.0% in 2010Q3, 5.9% in 2019Q4; last quarter -2.6% (s.d. 7.4)
- Real time is different: the filtered gap was only 1.4% in 2008Q3 and 0.3% in 2019Q4; s.d. of the revisions (smoothed minus filtered): 3.8 pp [Orphanides and van Norden \(2002\)](#)
- Model choice dominates: s.d. of the gap 4.5 (smooth trend), 0.53 (UC0), 5.4 (Hamilton); correlation smooth trend-Hamilton 0.65
- A Romanian output gap without its band and without the model behind it is not information

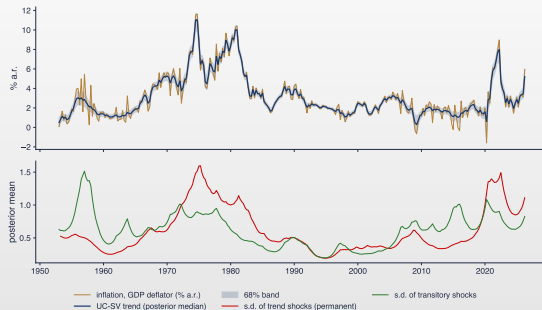
Recap: Trend-cycle decompositions

- BN, UC and Hamilton define the cycle differently; MNZ show that BN and UC agree once trend and cycle shocks may correlate
- In short, volatile samples (Romania) the correlation is not identified and the trend specification decides the gap
- Report filtered and smoothed estimates: the end-of-sample gap is the least reliable number

The UC-SV model of Stock and Watson (2007)

- $\pi_t = \tau_t + \sigma_{\eta,t}\eta_t$, $\tau_t = \tau_{t-1} + \sigma_{\varepsilon,t}\varepsilon_t$, $\ln \sigma_{j,t}^2 = \ln \sigma_{j,t-1}^2 + \nu_{j,t}$, $\nu_{j,t} \sim N(0, \gamma)$ Stock and Watson (2007)
 - ▶ π_t : inflation; τ_t : trend inflation (a random walk); $\sigma_{\eta,t}$, $\sigma_{\varepsilon,t}$: the time-varying s.d. of the transitory and trend shocks ($j \in \{\eta, \varepsilon\}$); $\eta_t, \varepsilon_t \sim N(0, 1)$
- $\gamma = 0.2$ is fixed, not estimated: the only tuning parameter; it governs how fast the volatilities can move
 - ▶ reduced form: IMA(1,1) with a time-varying MA coefficient θ_t , a function of $\sigma_{\varepsilon,t}/\sigma_{\eta,t}$
- Gibbs: $\tau_{1:n}$ by the precision sampler (time-varying weights); each $\ln \sigma_{j,1:n}^2$ by the KSC mixture on $\ln(\hat{\eta}_t^2 + c)$ and $\ln(\Delta\tau_t^2 + c)$
 - ▶ for Romania: three quarterly seasonal coefficients as an extra Gibbs block (weighted regression); the HICP is not seasonally adjusted
- Why it matters: the share of trend shocks decides how much of an inflation surge is permanent and how much a forecaster should extrapolate

US trend inflation, 1953–2026

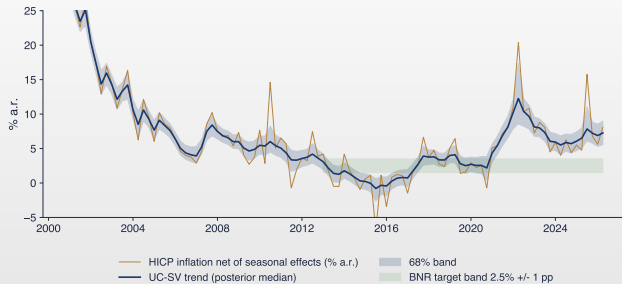


- GDP-deflator inflation, $T = 294$ quarters; UC-SV with $\gamma = 0.2$, Gibbs sampler; top: trend with 68% band; bottom: posterior means of $\sigma_{\varepsilon,t}$ (trend) and $\sigma_{\eta,t}$ (transitory)

Interpreting US trend inflation

- Trend: 8.9% in 1975Q1, 2.1% in 1995Q1, 1.4% in 2019Q4, 8.0% [6.6; 8.9] in 2022Q2, 5.2% [3.8; 6.0] in 2026Q2
- Trend shocks: s.d. 1.60 in 1975, 0.19 in 1995, 1.42 in 2022; transitory shocks: 0.88, 0.20, 0.90
- Implied MA coefficient θ_t : -0.20 (1975), -0.39 (1995), -0.37 (2019), -0.24 (2022): inflation became more like a random walk again in 2021–2022
- This is the forecasting lesson of [Stock and Watson \(2007\)](#): in calm periods smooth heavily; when trend volatility rises, a fixed-coefficient model reacts too slowly

Romanian trend inflation and the BNR target



- Quarterly HICP inflation (quarterly averages of the monthly index), $T = 102$, 2001Q1–2026Q2; UC-SV with $\gamma = 0.2$ and quarterly seasonal dummies; inflation-target band of the BNR since 2013

Interpreting Romanian trend inflation

- ▣ Trend: 8.4% when inflation targeting began (2005Q3), -0.8% in 2015Q3, 2.5% in 2019Q4, 8.1% in 2023Q1
- ▣ Last quarter: 7.3% [5.5; 9.0]; posterior probability that the trend lies inside 1.5–3.5%: 2%
- ▣ Since 2013 the trend has been inside the band only part of the time: below it in 2015–2016 (VAT cuts), above it since 2021
- ▣ Caveats: administered prices (energy caps, VAT changes) enter as trend shocks in a univariate model; a project can add them as regressors or as a separate component

Beyond: BSTS and regime switching

- Bayesian structural time series [Scott and Varian \(2014\)](#): local linear trend + seasonal + regression with a spike-and-slab prior on many regressors (Google Trends)
 - ▶ the same simulation smoother inside a Gibbs sampler; variable selection on the regression block
 - ▶ counterfactuals for policy evaluation (CausalImpact, [Brodersen et al. \(2015\)](#)): [Chapter 14](#)
- Discrete states: $s_t \in \{1, \dots, K\}$ with a Markov chain; the Hamilton filter is the Bayes filter with sums instead of integrals
 - ▶ state space models with regime switching (Kim filter, Gibbs sampling) [Kim and Nelson \(1999\)](#): [Chapter 7](#)
- Forecast evaluation of all these models (density forecasts, scoring rules): [Chapter 1](#)

Recap: Trend inflation

- ▣ UC-SV lets the split between permanent and transitory shocks change over time
- ▣ Two KSC blocks and one precision sampler make the Gibbs sampler fast
- ▣ US trend inflation rose sharply in 2021–2022; Romanian trend inflation is still above the target band

An open question

- How much of the 2021–2023 inflation surge was a shift in trend inflation, in the US and in Romania, and has it been reversed?
 - ▶ formal: the posterior of τ_t and of $\sigma_{\varepsilon,t}/\sigma_{\eta,t}$ in UC-SV; falsified if the trend has returned to its 2019 level with high posterior probability under every pre-registered specification
 - ▶ the univariate answer depends on one fixed parameter, γ : the mini-case below
- Why it matters: a central bank reacts differently to a transitory and to a persistent surge; anchoring is a statement about the trend, not about headline inflation
 - ▶ literature to start from: [Stock and Watson \(2007\)](#), [Cogley and Sargent \(2005\)](#), [Primiceri \(2005\)](#), [Morley, Nelson and Zivot \(2003\)](#)

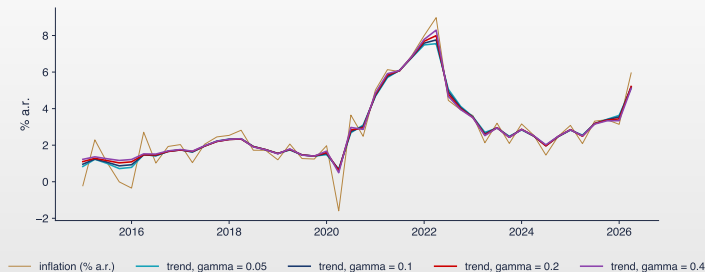
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers that estimate trend inflation with stochastic volatility for European countries after 2021; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Propose three observable implications that distinguish a trend shift from a sequence of large transitory shocks.
 - ▶ **code and replication:** ask for a UC-SV Gibbs sampler, then reproduce a known number first (the US trend of this lecture)
 - ▶ **robustness and critique:** Act as a hostile referee: which fixed choices of a UC-SV model could change the conclusion about 2022?
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The sampler targets the stated posterior: test it on a model with a known answer (the Kalman smoother, as in the three-sampler check)
- Convergence is reported: several chains, inefficiency factors, the effect of doubling the draws
- Fixed choices (γ , the KSC offset, priors, seasonal treatment, start date) are listed and varied before conclusions are drawn
- End-of-sample estimates are reported as filtered values with their bands, not as smoothed point estimates

Mini-case: one fixed parameter, four trends



- US UC-SV trend since 2015 for $\gamma \in \{0.05, 0.1, 0.2, 0.4\}$: trend in 2022Q2 7.6, 7.8, 8.0, 8.3% (inflation 9.0%); in the last quarter 5.1, 5.2, 5.2, 5.1%
- The ranking is stable but the level of the 2022 trend is not: an AI summary that reports “trend inflation peaked at X%” without γ hides a modelling choice

Project idea

- ▣ **Trend inflation and anchoring in Central and Eastern Europe:** replicate first, then extend
 - ▶ replicate: the US UC-SV trend of this lecture (2022Q2: 8.0%) and the Romanian trend (7.3% in 2026Q2)
 - ▶ extend: Romania, Hungary, Poland and Czechia; a multivariate UC-SV with a common euro-area trend; administered prices as a separate component; γ estimated with a prior instead of fixed
 - ▶ pre-register: data, sample, priors, the anchoring statistic (posterior probability inside the target band), the real-time exercise and the robustness grid
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ The Kalman filter is Gaussian conditioning applied recursively; the smoother, the likelihood and the simulation smoother follow from it
- ▣ Initialise nonstationary states exactly; treat zero variances as boundary problems
- ▣ Bayesian state space: sample whole paths; FFBS, DK and precision samplers are interchangeable
- ▣ Stochastic volatility is not GARCH; the KSC mixture and particle filters make it tractable
- ▣ Latent quantities (gaps, trends, time-varying slopes) come with model-dependent uncertainty: report bands, filtered values and alternatives

Self-assessment

Questions

- Why does a big-kappa likelihood fall without bound as κ grows?
- What does an ML estimate $\hat{\sigma}_\eta^2 = 0$ tell you about a local level?
- Why does the DK simulation smoother need only a mean smoother?
- Why is the EKF gain zero in the SV model?
- Why is $\ln \hat{L}$ biased although $\hat{L}$ is unbiased?
- Which restriction makes the UC cycle differ from the BN cycle?

Next:

- Regime-switching models
- Discrete hidden states: the Hamilton filter, the Kim smoother, EM and Gibbs for Markov switching, and state space models with regimes

Appendix: the Kalman filter from the lemma

- Given Y_{t-1} : $\begin{pmatrix} \alpha_t \\ y_t \end{pmatrix} \sim N \left(\begin{pmatrix} a_t \\ Z_t a_t \end{pmatrix}, \begin{pmatrix} P_t & P_t Z_t' \\ Z_t P_t & F_t \end{pmatrix} \right)$, $F_t = Z_t P_t Z_t' + H_t$
- Conditioning on y_t (equivalently on v_t , since $Y_t = (Y_{t-1}, v_t)$ and v_t is independent of Y_{t-1}) gives $a_{t|t}$ and $P_{t|t}$
- $\alpha_{t+1} = c_t + T_t \alpha_t + R_t \eta_t$ with η_t independent of Y_t : $a_{t+1} = c_t + T_t a_{t|t}$, $P_{t+1} = T_t P_{t|t} T_t' + R_t Q_t R_t'$
- Smoother: $\hat{\alpha}_t = a_t + \sum_{j=t}^n \text{Cov}(\alpha_t, v_j) F_j^{-1} v_j$ (the v_j are independent); $\text{Cov}(\alpha_t, v_j) = P_t L_t' \cdots L_{j-1}' Z_j'$ gives $\hat{\alpha}_t = a_t + P_t r_{t-1}$

Appendix: exactness of the Durbin–Koopman simulation smoother

- Gaussian model: $\alpha|y \sim N(\hat{\alpha}(y), V)$ with V independent of y , and $\hat{\alpha}(y)$ linear in y (affine with the intercepts)
- For a draw (α^+, y^+) from the joint law, $\alpha^+ - \hat{\alpha}(y^+) \sim N(0, V)$ and is independent of y^+ (and of y)
- Hence $\tilde{\alpha} = \hat{\alpha}(y) + \alpha^+ - \hat{\alpha}(y^+) \sim N(\hat{\alpha}(y), V)$; linearity gives $\hat{\alpha}(y) - \hat{\alpha}(y^+) = \hat{\alpha}_0(y - y^+)$, one smoother pass with zero intercepts
- Diffuse states: the exact diffuse smoother is invariant to the diffuse part of α_1^+ , so it can be set to a_1

Appendix: unbiasedness of the particle likelihood

- Bootstrap filter, one step: $\mathbb{E}[\frac{1}{N} \sum_i w_t^{(i)} | \mathcal{F}_{t-1}] = \int g(y_t|x) \hat{p}_N(x|Y_{t-1}) dx$, with $\hat{p}_N$ the empirical predictive law of the particles
- Telescoping the conditional expectations over $t = n, n-1, \dots, 1$ with multinomial or systematic resampling gives $\mathbb{E} \prod_t \hat{p}(y_t|Y_{t-1}) = p(y_{1:n})$ [Del Moral \(2004\)](#)
- Each factor is a ratio estimate, but the product is unbiased because resampling preserves the expected weights; normalised quantities (filtered means) are only consistent
- Jensen: $\mathbb{E} \ln \hat{L} \leq \ln \mathbb{E} \hat{L} = \ln L$; under a CLT for $\ln \hat{L}$ the bias is $-\frac{1}{2} \text{Var}(\ln \hat{L})$

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